Basic Linux Commands (Big Data Analysis Lab)

Linux Basics, Java Version & Environment Variables

Linux File & Directory Commands

- pwd – Show current working directory
- ls – List files and directories
- cd – Change directory
- mkdir – Create directory
- rmdir – Remove empty directory

Linux File Handling Commands

- touch file.txt – Create empty file
- cat file.txt – Display file content
- cp src dest – Copy files
- mv old new – Move or rename files
- rm file.txt – Delete file

Viewing & Searching Files

- head file.txt – View first lines
- tail file.txt – View last lines
- grep 'word' file.txt – Search text
- wc -l file.txt – Count lines
- sort file.txt – Sort file content

System & User Information Commands

- whoami – Current user
- uname -a – System information
- top – Running processes
- df -h – Disk usage
- free -h – Memory usage

Java Version Commands

- java -version – Check Java runtime version
- javac -version – Check Java compiler version
- which java – Java executable path
- readlink -f $(which java) – Actual Java location

Environment Variables Commands

- printenv – Show all environment variables
- env – Display environment variables
- echo $JAVA_HOME – Check Java path
- echo $PATH – Check system PATH

Setting Environment Variables

- export VAR=value – Temporary variable
- nano ~/.bashrc – Permanent setup
- source ~/.bashrc – Apply changes
- Used for JAVA_HOME, HADOOP_HOME

Conclusion

- Linux commands are essential for Big Data labs
- Java version & environment variables are mandatory for Hadoop
- These commands are important for lab exams and viva